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Project Charter

DMAIC project plan for customer-dispute documentation reliability

Project: Reducing Customer Dispute Case Documentation Rework
Department: Customer Protection Operations (hypothetical training scenario)
Project duration: 12 weeks | **Method:** DMAIC | **Status:** Proposed training project

Problem statement

During the most recent eight-week baseline period, 18% of completed customer-dispute cases were returned by Quality Assurance (QA) for missing, inaccurate, or inconsistent documentation. The average documentation cycle time was 42 minutes per case. Rework delays customer resolution, consumes specialist and QA capacity, and increases the risk of inconsistent regulatory and privacy records.

Business case

This project matters because documentation is both a customer-service output and a process control. Incorrect or incomplete records can delay dispute decisions, increase operating cost, weaken audit readiness, and create privacy or compliance exposure. Improving first-pass quality will make the process faster without lowering control standards.

Illustrative capacity impact: At 10,000 cases per month and 12 minutes of rework per returned case, an 18% return rate consumes about 360 staff hours monthly. Reducing the rate to 8% would free approximately 200 hours per month for value-added work.

SMART goal statement

Within 12 weeks, reduce the QA documentation return rate for the selected Customer Protection Operations unit from 18% to 8% or less and reduce mean documentation cycle time from 42 minutes to 32 minutes or less, while maintaining privacy-control compliance at or above 98.5%.

Improvement goals

QA documentation return rate	18%	8% or less	At least a 10-point absolute reduction
Mean documentation cycle time	42 minutes	32 minutes or less	At least a 10-minute reduction
Documentation defects per case	0.38	0.15 or less	At least a 60% reduction
Privacy-control compliance	98.5%	98.5% or higher	No degradation of the control guardrail

Training-data notice: The organization, baseline values, volumes, roles, and financial estimates in this charter are hypothetical planning assumptions created for this capstone. No Capital One or customer data are used.

Process description

The process begins when a customer dispute is assigned to the operations unit and ends when the disposition is approved, communicated, and closed. The selected process includes identity and privacy checks, information collection, investigation, formal case documentation, QA review, correction of documentation defects, final approval, and customer notification.

1	Receive and assign customer-dispute case	Intake system / supervisor	Correct queue and service clock
2	Verify identity, consent, and privacy requirements	Dispute specialist	Authorized access and complete controls
3	Confirm required information is available	Dispute specialist	No missing evidence or customer data
4	Gather transaction evidence and investigate	Dispute specialist	Accurate, traceable evidence
5	Document decision, rationale, actions, and controls	Dispute specialist	Complete and standardized record
6	Perform QA review	QA analyst	Consistent defect criteria
7	Correct defects when found	Dispute specialist	Rapid, complete rework
8	Approve outcome, notify customer, and close	QA / operations	Timely, compliant closure

Project scope

One U.S. customer-dispute operations unit; work from case assignment through QA approval; case-note templates; required-field logic; work instructions; QA defect codes; targeted specialist training; handoffs between specialist and QA.	Dispute eligibility policy; fraud-detection strategy; customer authentication platform; mobile-app design; vendor contracts; workforce headcount; other departments or international operations; final legal or regulatory policy changes.

Boundaries and constraints

The project may change documentation methods, controlled instructions, training content, and low-risk workflow validations. It may not change customer rights, decision authority, privacy policy, or dispute adjudication rules. The pilot will use de-identified case identifiers and existing approved systems.

Key stakeholders

Customers	Accurate and timely dispute resolution	High	Use complaint themes and cycle-time results as voice-of-customer inputs
Operations director	Capacity, cost, service, and risk	High	Sponsor reviews at Define, Analyze, Pilot, and Close
Dispute specialists	Usable instructions and efficient workflow	High	Workshops, pilot feedback, and training
QA team	Consistent first-pass quality and defect coding	High	Definition alignment and weekly defect review
Privacy and risk partners	Control effectiveness and auditability	High	Approve controls and monitor guardrails
Product / technology	Feasible system validations	Medium	Review requirements and support pilot configuration
Training / knowledge management	Controlled learning content	Medium	Coauthor work instructions and sustainment plan

Project team

Executive Sponsor - Director, Customer Protection Operations	Remove barriers, approve scope and resources	Charter, pilot, and closure approval
Process Owner - Dispute Operations Manager	Own process performance and sustainment	Operational process decisions
Yellow Belt Project Lead - Mary Langley, Process Documentation Specialist	Coordinate DMAIC work, documentation, analysis, and reporting	Day-to-day project coordination
QA Analyst	Define defects, validate data, and review results	QA criteria recommendations
Dispute Specialist SME	Represent frontline workflow and test solutions	Usability recommendations
Data Analyst	Sampling, data quality, and statistical testing	Analysis-method recommendations
Privacy / Risk Partner	Validate privacy and process controls	Control approval or escalation

Twelve-week timeline

DEFINE - charter, VOC, CTQs	X	X										
MEASURE - definitions, MSA, baseline			X	X	X							
ANALYZE - Pareto, causes, hypotheses						X						
IMPROVE - solution design							X					
IMPROVE - pilot and training								X	X			
MEASURE - post-pilot sample										X		
CONTROL - controls and handoff											X	
CLOSE - results and lessons learned												X

Major deliverables and gates

Define approved	2	Signed charter, CTQs, stakeholder review
Measure complete	5	Operational definitions, MSA, baseline sample
Root causes validated	6	Pareto, process analysis, cause evidence
Pilot approved	7	Solution design, risk review, training plan
Control handoff	11	Control plan, owner acceptance, monitoring dashboard
Project close	12	Before/after results, lessons learned, final report

Key risks and mitigations

Inconsistent QA interpretation	Calibrate reviewers on 30 common cases; target at least 90% agreement and kappa of 0.80 or higher.
Pilot increases handling time	Use a two-week pilot and monitor daily; retain a rollback version of instructions and validation rules.
Privacy data are exposed	Use approved systems, least-privilege access, and de-identified analysis extracts.

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Team Charter

Working agreements for the customer-dispute documentation improvement team

Team purpose: Work collaboratively to reduce documentation rework and cycle time while protecting customer privacy, regulatory obligations, and the accuracy of dispute decisions. These expectations will be reviewed, revised if needed, and approved by the full team at kickoff.

Team membership and responsibilities

Sponsor	Strategic direction and barrier removal	Approves scope, resources, pilot, and close
Process Owner	Operational leadership	Accepts controls and owns sustained performance
Yellow Belt Project Lead	Facilitation, analysis, documentation, coordination	Maintains charter, action log, decisions, and deliverables
QA Analyst	Defect knowledge and audit perspective	Maintains defect definitions and QA calibration
Dispute Specialist SME	Frontline expertise and usability feedback	Tests new workflow and represents specialists
Data Analyst	Sampling and statistical support	Validates data quality and hypothesis tests
Privacy / Risk Partner	Control and policy oversight	Approves privacy safeguards and escalates unacceptable risk

Shared expectations

1. Customer and control focus	Evaluate changes against customer timeliness, record accuracy, and privacy requirements.
2. Evidence before opinion	Use operational definitions and data to validate causes; do not assume the employee is the problem.
3. Attendance and preparation	Attend scheduled meetings, review materials beforehand, and notify the lead when a delegate is needed.
4. Respectful participation	Invite differing viewpoints, critique the process rather than the person, and avoid interrupting.
5. Reliable commitments	Accept actions only with an owner and due date; raise risks before deadlines are missed.
6. Scope discipline	Place out-of-scope ideas in the parking lot and request sponsor approval before expanding boundaries.
7. Data integrity	Do not alter, omit, or selectively report records; document exclusions and limitations.
8. Confidentiality	Use approved systems and de-identified data; never place customer information in project presentations.

Meeting and communication plan

Core-team meeting	Weekly, 45 minutes	Project lead and core team	Status, decisions, risks, and action log
Sponsor checkpoint	Every two weeks	Sponsor, process owner, project lead	Gate decision and barrier removal
Pilot huddle	15 minutes daily during Weeks 8-9	Lead, SME, QA, supervisor	Pilot issues and same-day response
Working sessions	As needed	Relevant subject-matter experts	Process map, definitions, analysis, or solution design

Decision-making and escalation

The team will seek consensus after reviewing evidence and stakeholder impact. When consensus is not possible, the Process Owner decides operational matters, the Privacy / Risk Partner decides control acceptability, and the Sponsor decides scope, funding, or unresolved cross-functional issues. Urgent customer or compliance risks are escalated immediately rather than held for the next meeting.

Conflict-resolution approach

1	State the disagreement in neutral process language and identify the decision needed.
2	Review the charter, operational definitions, data, and customer/control requirements.
3	Test a small, reversible option when evidence is incomplete.
4	Escalate to the assigned decision owner if the issue cannot be resolved within two business days.

Definition of done

☐ Deliverable reviewed	Appropriate SME and control partner comments resolved
☐ Data verified	Source, exclusions, sample size, and calculations documented
☐ Version controlled	Owner, approval date, and next review date recorded
☐ Sustaining owner assigned	Process Owner accepts monitoring and response plan
☐ Results communicated	Affected employees receive clear instructions and training

Team participation and approval

Executive Sponsor	_____	_____
Process Owner	_____	_____
Project Lead	Mary Langley	_____
Core Team Representative	_____	_____

Data Collection Plan

Measures, operational definitions, sampling, ownership, and data quality

Sampling plan: Collect a stratified random sample of 200 completed cases during the four-week baseline and 200 cases after the pilot. Preserve the same weekday, shift, work type, and specialist-mix proportions. Analyze de-identified case IDs only.

Measures and data types

QA documentation return	Case sent back by QA for at least one documentation defect before final approval	Attribute - binary	Yes/No; percent returned	QA workflow status	All sampled cases; weekly summary	QA Analyst	18% / 8% or less
Documentation cycle time	Minutes from case assignment to first QA submission; exclude documented customer-wait time	Continuous	Minutes; mean and median	System timestamps	All sampled cases; weekly summary	Data Analyst	42 / 32 minutes or less
Defects per case	Count of distinct QA documentation defects using the approved defect code list	Discrete count	Count; mean per case	QA defect log	Every sampled case	QA Analyst	0.38 / 0.15 or less
Privacy-control compliance	All required identity, consent, access, and disclosure checks documented correctly	Attribute - binary	Pass/Fail; percent passed	Privacy checklist / audit	100% of pilot sample	Risk Partner	98.5% / 98.5% or higher
Primary defect category	Single category representing the earliest or most consequential documentation defect	Nominal categorical	Category frequency / percent	QA defect log	Each returned case	QA Analyst	Diagnostic; no numeric target

Planned defect categories

Missing required field	Identity check, evidence source, approval, customer communication	Required element is blank or absent
Incorrect information	Wrong amount, date, transaction, disposition, or customer detail	Recorded value conflicts with verified source
Insufficient rationale	Decision or action lacks supporting explanation	Reviewer cannot reproduce reasoning from the record
Format / standardization	Unapproved abbreviation, duplicate note, incorrect template section	Record violates controlled documentation standard
Privacy / control	Missing consent, inappropriate data exposure, access issue	Any required privacy-control check fails

Collection procedure

1	Create the eligible case list for the defined unit and period.	Exclude only cases meeting documented exclusion rules.
2	Stratify by shift, weekday, dispute type, and specialist tenure band.	Use the same proportions before and after the pilot.
3	Select 200 cases using a reproducible random-number method.	Retain the sample seed and selection log.
4	Extract assignment and first-QA-submission timestamps.	Check for missing, reversed, or duplicate timestamps.
5	QA reviewer records return status, defect count, and category.	Use controlled definitions and one primary category.
6	Risk reviewer records privacy-control pass/fail.	Escalate any actual control failure immediately.
7	Data Analyst validates, locks, and summarizes the dataset.	Document exclusions; no undocumented data cleaning.
8	Repeat the same method after the pilot.	Compare equivalent before/after populations.

Measurement-system checks

QA agreement	Two reviewers independently classify 30 cases	At least 90% agreement and kappa 0.80 or higher	Clarify definitions, retrain, and repeat
Timestamp accuracy	Compare 20 extracts with source-system history	100% match for start/end timestamps	Correct query or field mapping
Defect-count consistency	Recount 30 cases after one week	At least 95% exact agreement	Review defect boundaries and examples
Sampling comparability	Compare before/after strata	No material shift in dispute type, shift, or tenure mix	Reweight or redraw sample

Data governance and storage

The working dataset will contain only a project-specific random case key, dates, durations, outcome flags, counts, categories, and sampling strata. Names, account numbers, full transaction descriptions, and customer messages will not be copied into the analysis file. Access is limited to the project team, changes are version controlled, and the Process Owner approves retention and disposal.

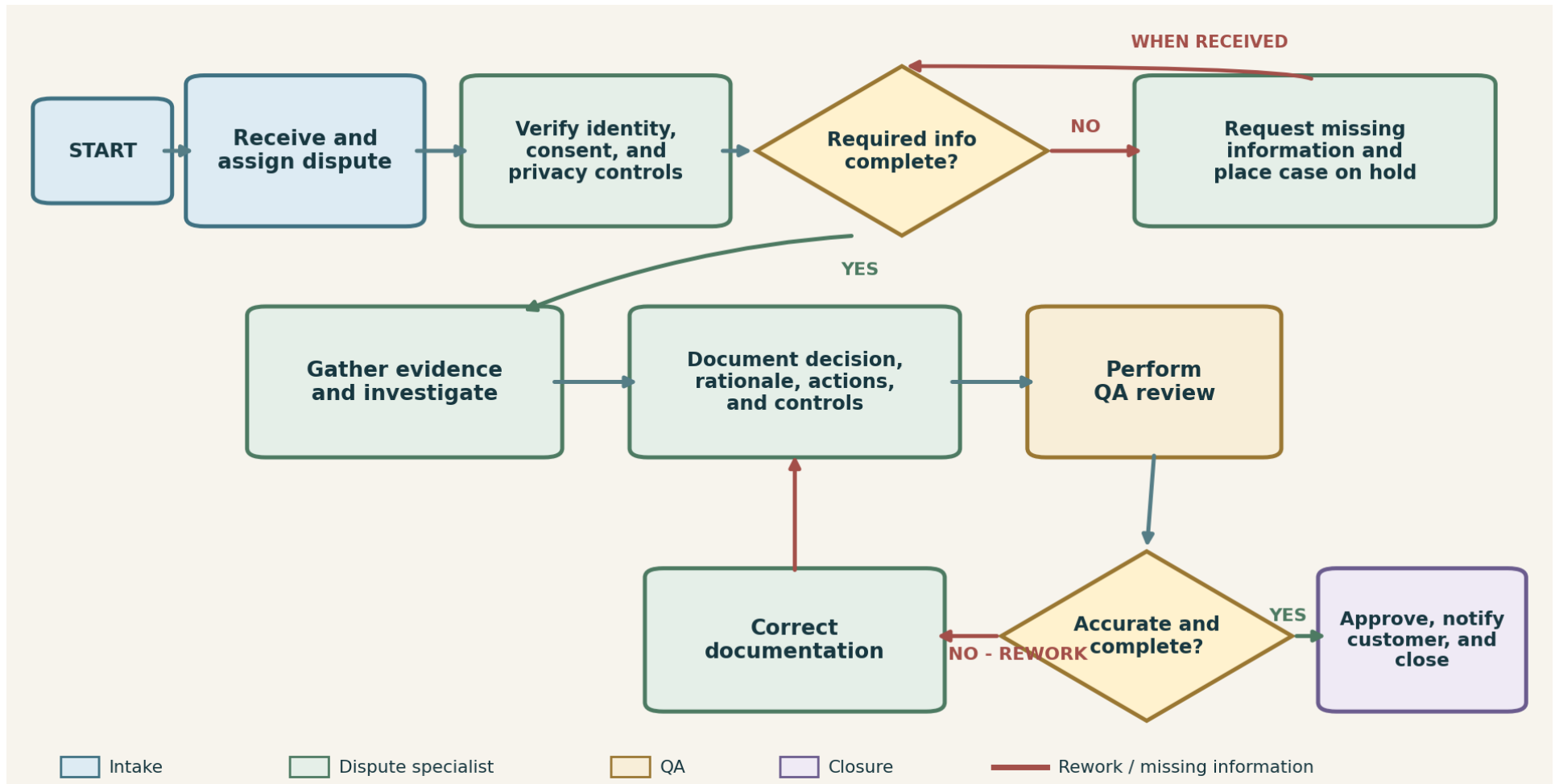
Analysis plan

QA return rate	Run chart, proportion, confidence interval	One-sided two-proportion test
Cycle time	Histogram, mean, median, standard deviation	One-sided two-sample t-test; nonparametric check if strongly skewed
Defects per case	Frequency plot, mean, variance	Poisson or negative-binomial rate comparison
Privacy compliance	Pass rate and confidence interval	One-sided noninferiority proportion test
Defect category	Pareto chart and category proportions	Chi-square test of homogeneity

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Process Map

Current-state customer-dispute documentation workflow and rework loops



Primary improvement opportunities: standardize the documentation template; add required-field and part-number-style validations; clarify work instructions with examples; calibrate QA defect definitions; and deliver focused training before the pilot.

Null and Alternative Hypotheses

Before-and-after tests for each proposed project measure

Comparison: Baseline sample (before improvement) versus post-pilot sample (after improvement). Use a one-sided significance level of $\alpha = 0.05$ for directional improvement tests. Privacy compliance is evaluated as a noninferiority guardrail.

Notation

p(before), p(after)	Population proportions before and after improvement
μ (before), μ (after)	Population means before and after improvement
λ (before), λ (after)	Mean documentation-defect counts before and after improvement

1. QA documentation return rate

Measure	Proportion of sampled cases returned by QA for one or more documentation defects.
Null hypothesis (H0)	H0: $p(\text{after}) \geq p(\text{before})$. The return rate did not decrease.
Alternative hypothesis (Ha)	Ha: $p(\text{after}) < p(\text{before})$. The return rate decreased after improvement.
Suggested test	One-sided two-proportion z-test.
Practical success	Statistical evidence of reduction and a post-pilot rate of 8% or less.

2. Documentation cycle time

Measure	Mean minutes from assignment to first QA submission, excluding documented customer-wait time.
Null hypothesis (H0)	H0: $\mu(\text{after}) \geq \mu(\text{before})$. Mean cycle time did not decrease.
Alternative hypothesis (Ha)	Ha: $\mu(\text{after}) < \mu(\text{before})$. Mean cycle time decreased after improvement.
Suggested test	One-sided independent-samples t-test; verify assumptions and use a Mann-Whitney check if needed.
Practical success	Statistical evidence of reduction and a post-pilot mean of 32 minutes or less.

3. Documentation defects per case

Measure	Mean count of distinct documentation defects recorded for each sampled case.
Null hypothesis (H0)	H0: $\lambda(\text{after}) \geq \lambda(\text{before})$. The mean defect count did not decrease.
Alternative hypothesis (Ha)	Ha: $\lambda(\text{after}) < \lambda(\text{before})$. The mean defect count decreased after improvement.
Suggested test	One-sided Poisson rate comparison; use a negative-binomial model if overdispersion is present.
Practical success	Statistical evidence of reduction and a post-pilot mean of 0.15 defects per case or less.

4. Privacy-control compliance

Measure	Proportion of sampled cases passing all required privacy-control checks.
Null hypothesis (H0)	H0: $p(\text{after}) \leq p(\text{before}) - 0.01$. Compliance is worse by at least one percentage point.
Alternative hypothesis (Ha)	Ha: $p(\text{after}) > p(\text{before}) - 0.01$. Compliance is noninferior within a one-point margin.
Suggested test	One-sided noninferiority test for two proportions.
Practical success	Post-pilot compliance remains at or above 98.5%, with no confirmed increase in privacy risk.

5. Primary defect-category distribution

Measure	Relative frequencies of missing field, incorrect information, insufficient rationale, format, and privacy/control defects.
Null hypothesis (H0)	H0: The before-and-after defect-category distributions are the same.
Alternative hypothesis (Ha)	Ha: At least one defect-category proportion differs after improvement.
Suggested test	Chi-square test of homogeneity; use Fisher's exact method if expected counts are too small.
Practical success	The targeted categories decline and the Pareto chart no longer shows the same dominant causes.

Decision and interpretation rules

p-value < 0.05 and practical target met	Reject H0; evidence supports a meaningful improvement.
p-value < 0.05 but practical target missed	A change is detected, but improvement is not yet sufficient.
p-value ≥ 0.05	Do not reject H0; the sample does not provide enough evidence of improvement.
Guardrail fails	Do not standardize the change until the privacy/control issue is corrected.

Statistical significance will not be used alone. Confidence intervals, effect sizes, data quality, customer impact, process stability, and the defined success targets will all be considered before recommending full implementation.

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Yellow Belt and Capstone Reflection

Lessons learned from applying Six Sigma thinking to a documentation process

What the training changed for me

The Yellow Belt training gave me a structured way to approach problems that I had often addressed through intuition and experience. I naturally notice confusing instructions, unnecessary steps, repeated work, and breakdowns between people and systems. Six Sigma showed me how to turn those observations into a disciplined improvement project. Instead of beginning with a preferred solution, I learned to define the problem, identify the customer requirement, establish a baseline, and test whether a change actually improved performance.

The project charter was one of the most useful tools because it forces the problem to become concise, measurable, and bounded. A broad statement such as 'documentation needs improvement' does not tell a team where to work or when it has succeeded. In this capstone, defining an 18% QA return rate, a 42-minute cycle time, a specific operations unit, and a 12-week period created a project that could realistically be managed. The scope section was equally important because process-improvement efforts can easily expand into policy, technology, staffing, and unrelated departments.

Measurement before improvement

Another major lesson was that data are only useful when everyone measures the same thing. Terms such as defect, rework, cycle time, and complete documentation can appear obvious while different reviewers interpret them differently. Operational definitions, sampling rules, and measurement-system checks protect the project from drawing conclusions from inconsistent data. I also learned that data type matters. A yes-or-no return decision is attribute data, time is continuous data, a defect count is discrete data, and defect category is nominal data. Each type requires a different summary and may require a different statistical test.

Most important realization: Documentation is not merely a record of the process. In a regulated environment, clear documentation, controlled instructions, and reliable templates are part of the process control system itself.

Learning from the capstone

Planning the customer-dispute documentation project helped connect the tools from the course. The process map made the rework loops visible. An affinity diagram or cause-and-effect analysis could organize possible causes without immediately blaming the worker. A Pareto chart could then show which documented defect categories account for the largest share of returns. Histograms and descriptive statistics would show whether cycle time is consistent or contains a long tail. Finally, the hypotheses define what evidence would be needed to decide whether the pilot produced more than random variation.

The capstone also reinforced the importance of balancing improvement measures with guardrails. Reducing cycle time would not be a success if privacy compliance declined or if specialists began omitting important rationale. The project therefore includes a privacy-control measure, clear decision authority, and a requirement to stop or correct the pilot if the guardrail fails. That balance is especially important in financial services, where speed, customer experience, documentation quality, and compliance must work together.

Connection to my professional work

My background in technical documentation, structured content, work instructions, and regulated insurance gives me a strong foundation for this type of work. I understand that a technically correct instruction can still fail if it is difficult to follow, poorly organized, outdated, or disconnected from the actual workflow. I also understand the need for privacy, consumer protection, auditability, and escalation in financial processes. Six Sigma adds a measurement and improvement framework to those documentation skills.

Going forward, I can use DMAIC to make my documentation projects more valuable. I can define the business problem behind a document, identify the critical customer requirements, measure errors or delays, analyze their causes, pilot improvements, and create controls for maintenance. This approach will help me speak the language of operations, quality, risk, product, and leadership while still contributing my strengths as a technical writer and process-documentation specialist.

Closing reflection

The Yellow Belt training taught me that continuous improvement is not a single correction or a one-time rewrite. It is a repeatable way of learning from a process. The capstone showed me how clear writing, stakeholder participation, reliable data, and statistical thinking can work together to produce a solution that is practical, measurable, and sustainable.

Personal takeaway: My technical-writing mindset and Six Sigma thinking reinforce one another: both require curiosity, precision, evidence, empathy for the user, and the willingness to revise a system until it works reliably.